Activity 1

Compare india and other 4 Nation of basis of E-Waste

→	Country	Annual Generation Volume	Per capital Generation	Documented formal Recycling Rate	Regulatory and operational framework	Primary challenge adv.
①	India	3.2 - 4.0 Mt/year	3.1-3.3 kg/person (Low)	~10-15% formal (~90%) handle informally	Ewaste rules EPR Portal and Credit System	Domainant informal sector using hazardous extraction methods.
②	China	~10-12+ Mt/year (Highest)	~9-10 kg/person (Med.)	~20-25% formal	WEEE Recycling Regulation and State-subsidized recycling funds	

1] Country

India

↳ Annual Generation volume is 3.2 - 4.0 Mt/year

↳ 3.1 - 3.3 kg/person (Low) per capital Generation

↳ ~10-15% formal and ~90% informally recycling rate

↳ Domainant informal sector using hazardous extraction method

2] ~~10-12~~ China

↳ ~10-12+Mt/year as annual Generation volume

↳ ~9-10 kg/person (Medium)

↳ ~20-25% formal Documented Recycling Rate

↳ WEEE Recycling Regulation and Subsidized recycled funds

3] United States

~7.0-7.2Mt year as annual Generation volume

~21-23 kg/person (Very High) Per capital Generation

~15-20% formal recycling rate

↳ ~~Documented Recycling~~ State level legislation (No single federal mandatory EPR frame'work)

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c) Exceptionally high per capital rates; reliance on exporting untreat e-waste

4] Germany

~1.6 MT/year

~19-20 kg/person (very High)

~45-50% formal

- EU WEEE Directive and German Electrical and electronic Equipment Act (ElektroG)
- Adv — Highly stucted, mandatory retail takeback and municipal collection networks.

5) Japan

→ ~2.5 MT/year

→ ~20 kg/person (very High)

→ ~30-35% formal

→ Home application Recycling law and Small appliance Recycling Act

→ Adv

High consumer compliance and manufactures-backed integrated recycling pipeline

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Activity 3

5 way to tackle EWaste

1) Formalize the informal Sector
Integrate india's 1M+ informal informal workers into licensed collection networks, equipping them with personal protective gear to safely dismantle waste rather than using open burning or acid baths

2) Adopt Green Recycling Technologies
Shift away from toxic pyrometallurgy and cyanide leaching by adopting biometallurgy and natural clay nano traps to capture toxic heavy metals..

3) Enforce EPR Policies: Strictly enforce national e-waste rules and producer recycling traget to convert policy into actual collecti-

4) Promote Urban Mining - Build recovery hubs for circuit boards

5) Stop Landfilling and Burning
Ban electronic in regular landfills and eliminate open burning to prevent heavy metal leaching and toxic dioxins

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